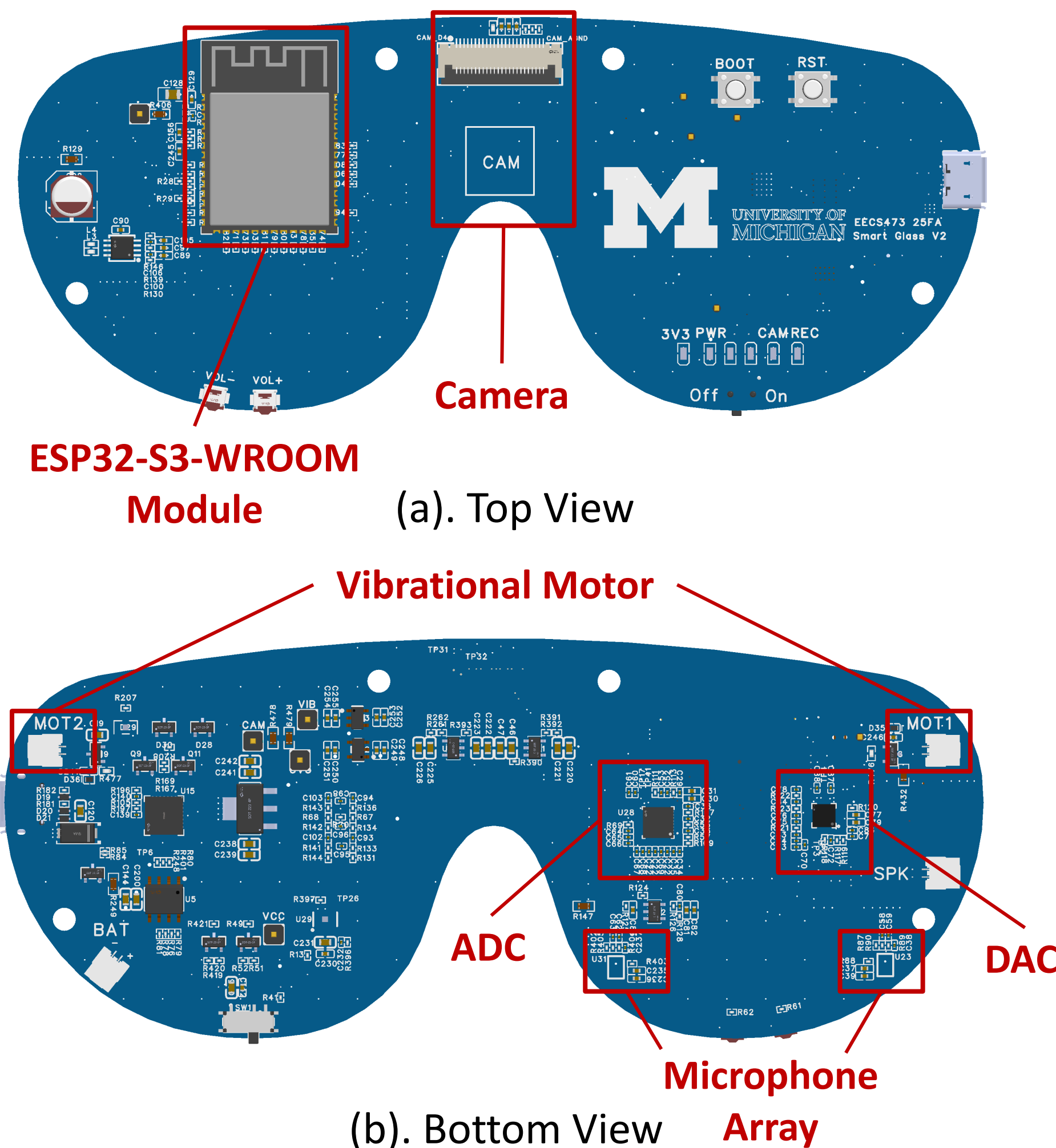


Introduction

- A system built to assist **visually impaired people** in **navigation** and **scene description**.
- Low-cost**: Uses commodity hardware and compact models so the system costs less than \$300, making it accessible to everyday users.
- Real-time**: Processes sensor and camera data on-device to deliver instant audio and haptic feedback for safe navigation and scene description.
- Privacy**: All perception and language models run locally and offline, so no video or audio ever leaves the device.

Smart Glass System

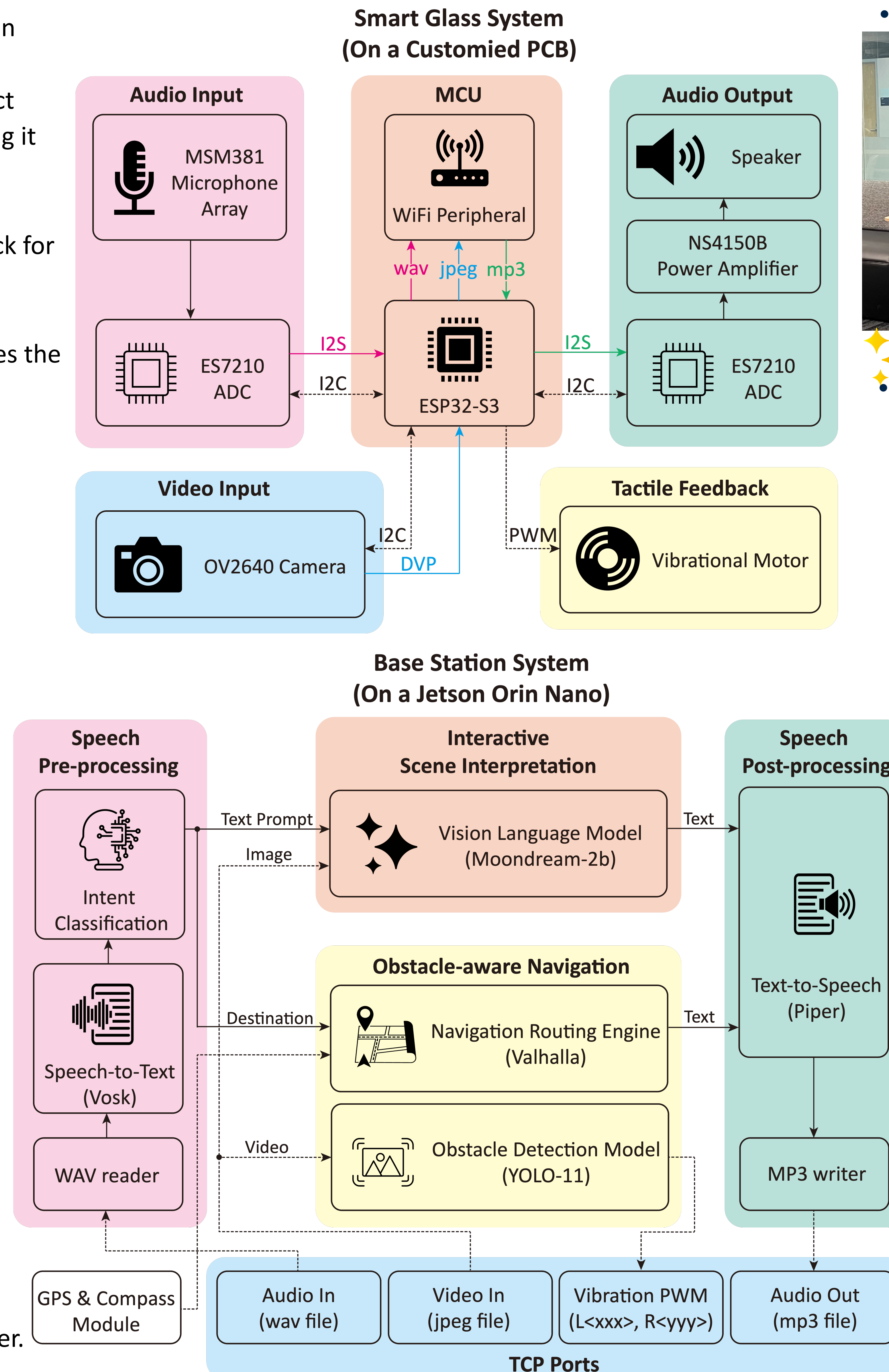
- Customized PCB



Software Overview

- Task 1: Wake word ("Hi ESP") detection.
- Task 2: Audio recording and streaming.
- Task 3: Take video from camera and stream out.
- Task 4: Receive audio and play on the speaker.
- Task 5: Vibrational motor PWM receiver and driver.

Function Block Diagram



Results

Scene Interpretation

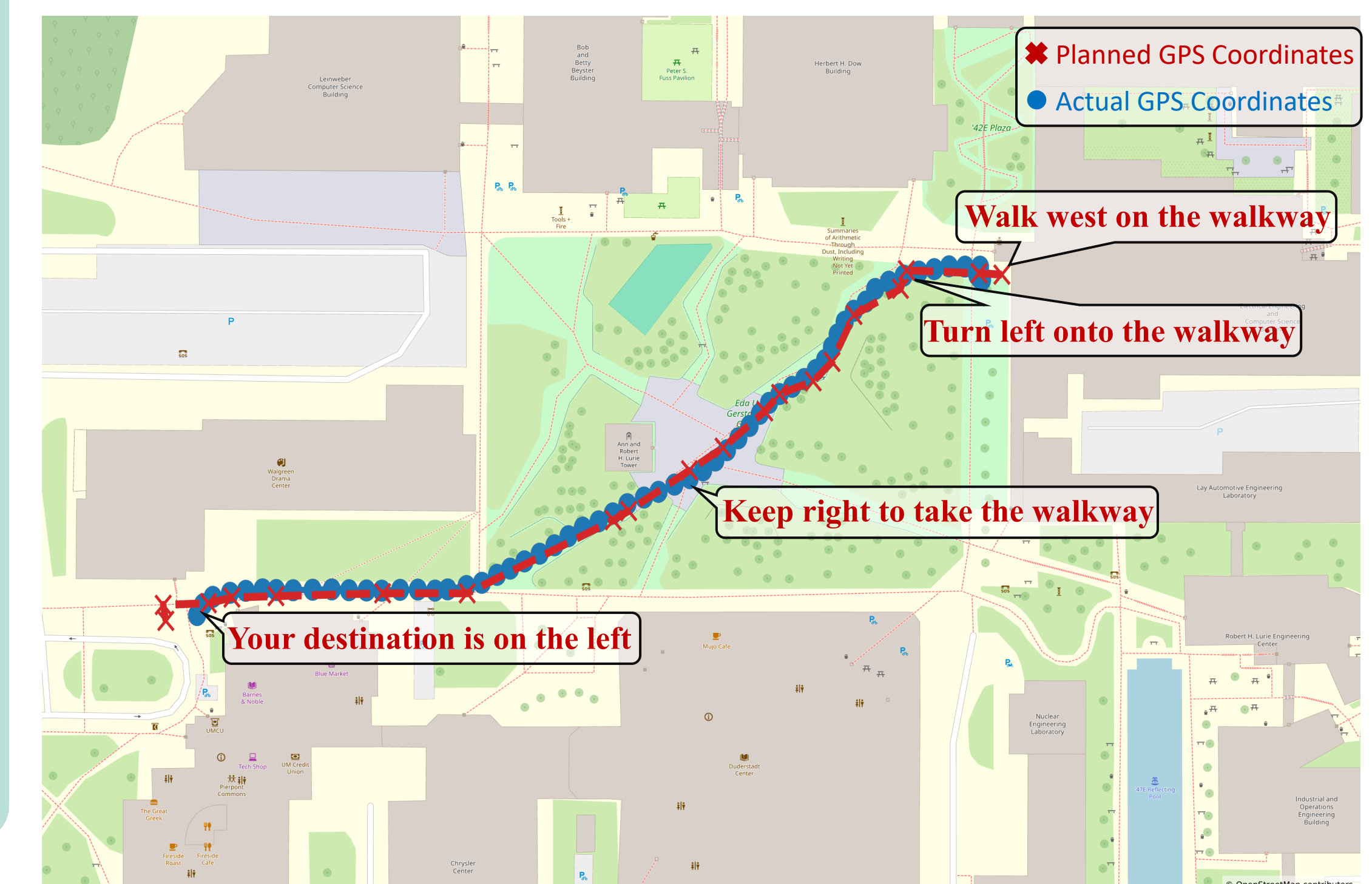


What's in front of me?

Obstacle-aware Navigation



Vibrational Motor Feedback



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